

Unit 14, Lesson 3: Two-Way Tables and Segmented Bar Graphs



Benchmarks

MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.



Learning Targets

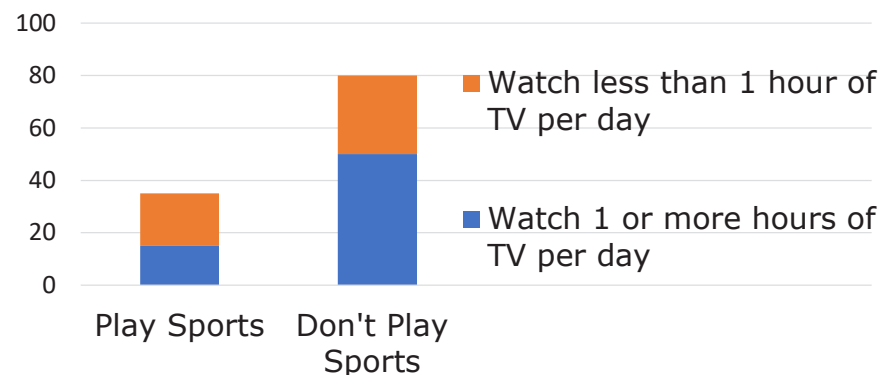
- ☐ I can correctly match a segmented bar graph and data in a two-way table.
- ☐ I can interpret a graphical display for bivariate categorical data.



14.3.1 Warm-Up: Notice and Wonder

A segmented bar graph is shown.

TV Watched by People Who Play Sports and Those Who Don't



1. What do you notice about the display?
2. What do you wonder?



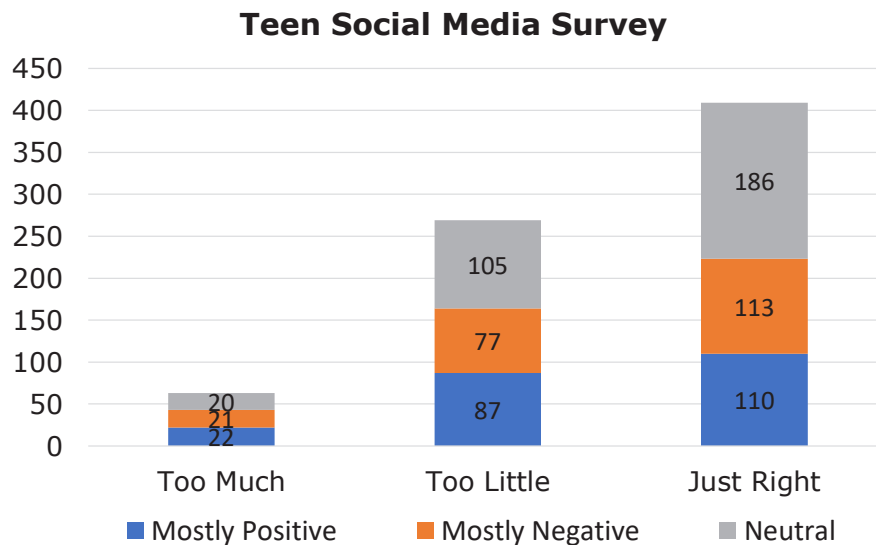
14.3.2 Exploration: Segmented Bar Graphs

Bivariate categorical data can be displayed in a two-way frequency table as shown:

Teen Social Media Survey

		Effect on People Your Age			
Time spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

It can also be displayed in a **segmented bar graph**.



Three two-way frequency tables and four segmented bar graphs are shown.

Table 1

Peoples' Social Media Preference by Age Group

	Less Than 18 Years Old	18 – 30 Years Old	Over 30 Years Old	Total
Snapchat	147	72	90	309
Facebook	23	56	112	191
Total	170	128	202	500

Table 2

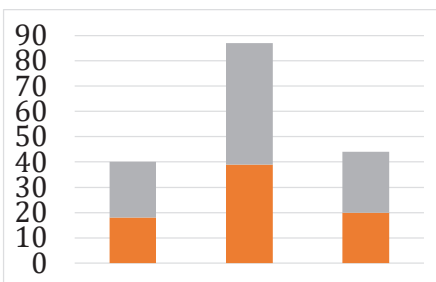
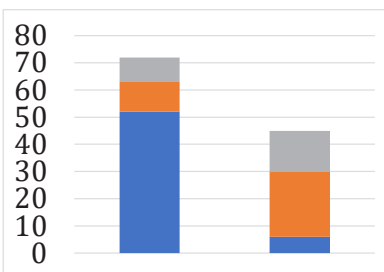
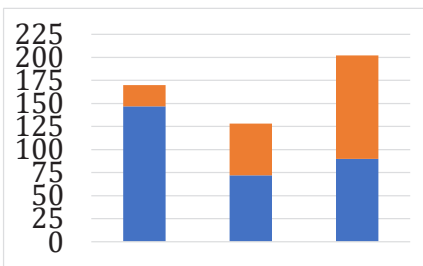
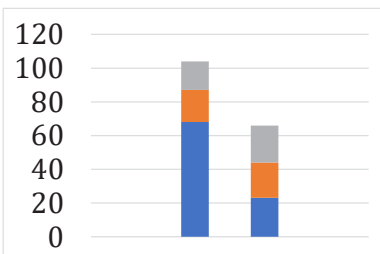
Points Scored During a Volleyball Match

	University of Miami, Florida	Jacksonville University, Florida	Total
Spike Kills	18	22	40
Service Aces	39	48	87
Other	20	24	44
Total	77	94	171

Table 3

Emotional States Related to Meditation Practice

	Meditated	Did Not Meditate	Total
Calm	52	6	58
Excited	11	24	35
Agitated	9	15	24
Total	72	45	117

Graph A**Graph B****Graph C****Graph D**

1. Work with your partner to match each two-way frequency table to the segmented bar graph that displays the same data.

Two-Way Frequency Table	Segmented Bar Graph
1	☐ A ☐ B ☐ C ☐ D
2	☐ A ☐ B ☐ C ☐ D
3	☐ A ☐ B ☐ C ☐ D

2. Work with your partner to label the bars and their sections in the segmented bar graphs so that the categories represented by each bar are indicated.
3. Discuss with your partner how joint frequencies are displayed in a segmented bar graph. Summarize your discussion.



14.3.3 Bring It Together: Representing Bivariate Categorical Data With Segmented Bar Graphs

Another way to represent the bivariate categorical data in a two-way frequency table is with a segmented bar graph.

1. In a two-way frequency table, marginal frequencies are found in the

☐ center
☐ margins

 of the table. These numbers

express the

☐ totals
☐ parts

 of each category.

2. Carolyn concluded that marginal frequencies are represented by the height of each bar on a segmented bar graph.

Bailey argued that the marginal frequencies are represented by the sum of the segment heights with the same color for each color.

Which student do you agree with? Explain your reasoning.

3. In a two-way frequency table, joint frequencies are found in the

☐ center
☐ margins

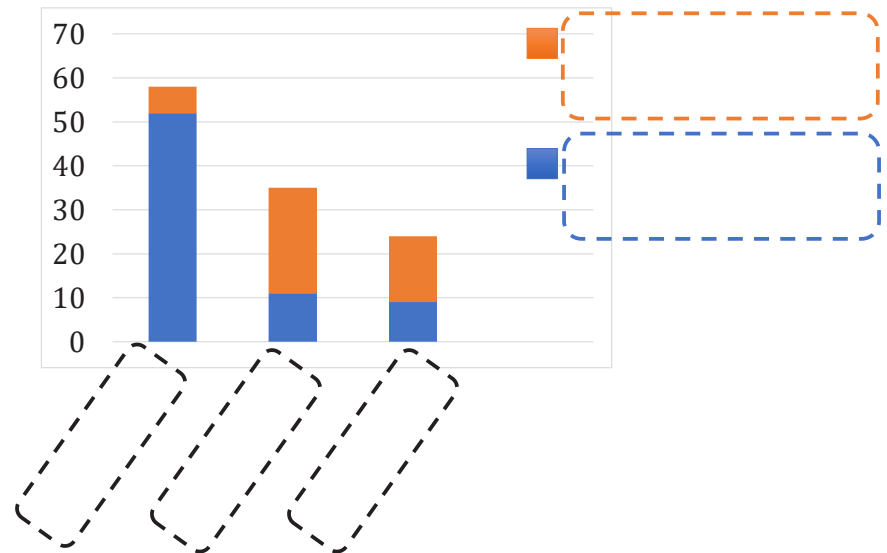
 of the table. Joint frequencies are represented by the

- ☐ total height of a bar.
☐ segments of different colors within each bar.

4. Consider the data in the table and segmented bar graph.

People's Emotional States Related to Meditation Practice

	Meditated	Do not Meditate	Total
Calm	52	6	58
Excited	11	24	35
Agitated	9	15	24
Total	72	45	117



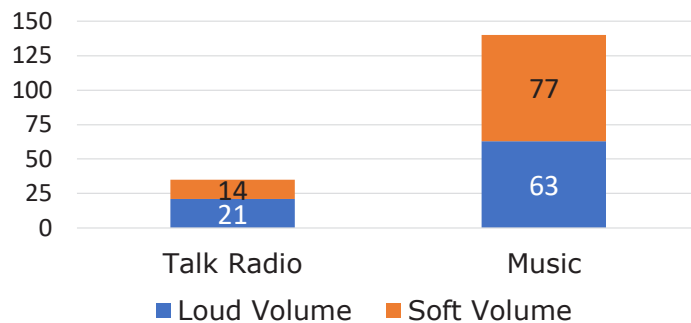
- a. Circle the marginal frequencies in the table.
- b. Label each marginal frequency in the dotted boxes.
- c. Label the value for each joint frequency.



14.3.4 Try It: Interpreting Segmented Bar Graphs

- Long-haul truckers were recently surveyed at a rest area in Tallahassee, Florida. The truckers were asked about what and how they primarily listen to audio in their trucks while on the road.

Long-Haul Truck Driver Listening Preferences



Determine whether each of the following statements is true or false. If the statement is false, correct the statement to make it true.

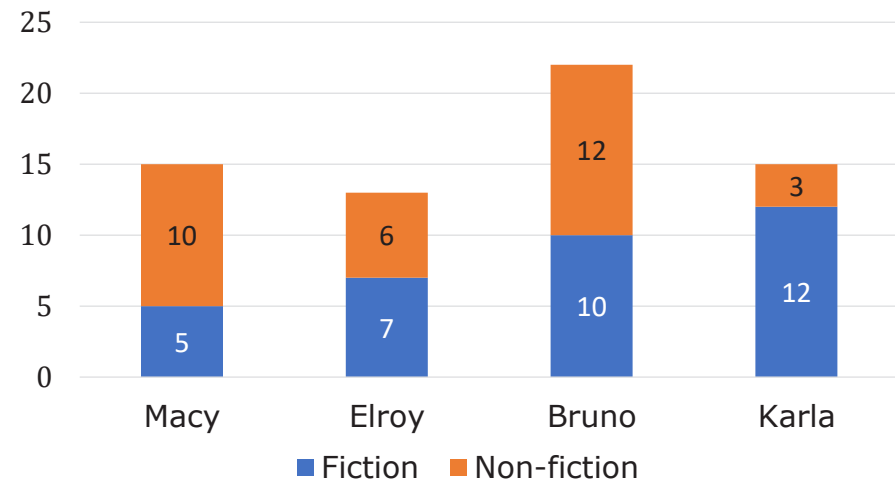
14 of those surveyed prefer loud volume.	☐ True ☐ False
140 of those surveyed prefer soft volume.	☐ True ☐ False
63 of those surveyed prefer music at a soft volume.	☐ True ☐ False
More people prefer music than talk radio.	☐ True ☐ False
More people prefer talk radio to be at a loud volume than at a soft volume.	☐ True ☐ False



14.3.5 Wrap-Up: Ticket Out the Door

Macy, Elroy, Bruno, and Karla each kept track of the books they read for a year. Their data is represented in the segmented bar graph shown. Use the segmented bar graph to answer the questions.

Books Read



- What does the value 7 mean in this context?
- How many fiction books were read?